

Introduction

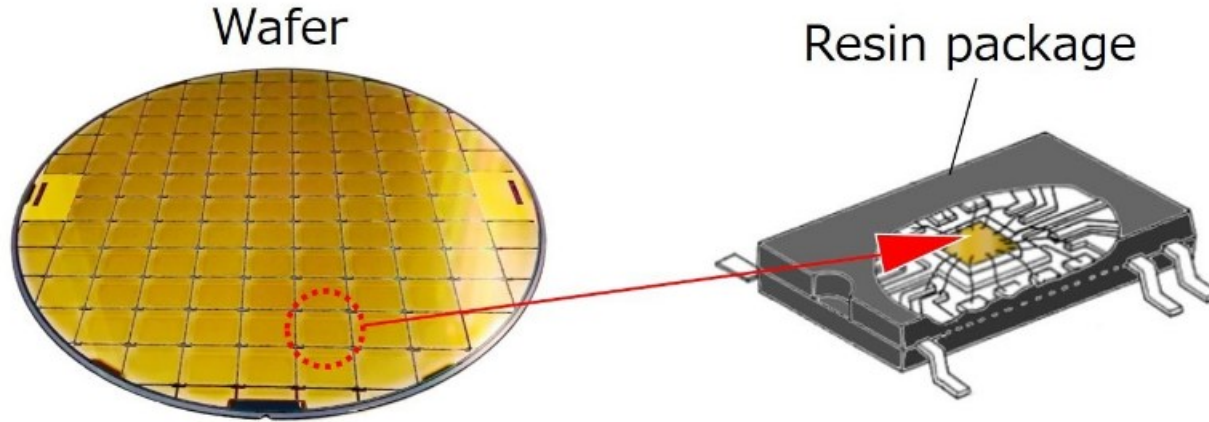
EE 350 – Electronics II

There are two main types of electronic circuits:

- 1) Analog Circuits
- 2) Digital Circuits
- 3) Mixed-Signal (combination of analog and digital circuits)

The circuits can be implemented at both PCB (Printed Circuit Board) or semiconductor (integrated circuit [IC] design, Si, GaAs) level.

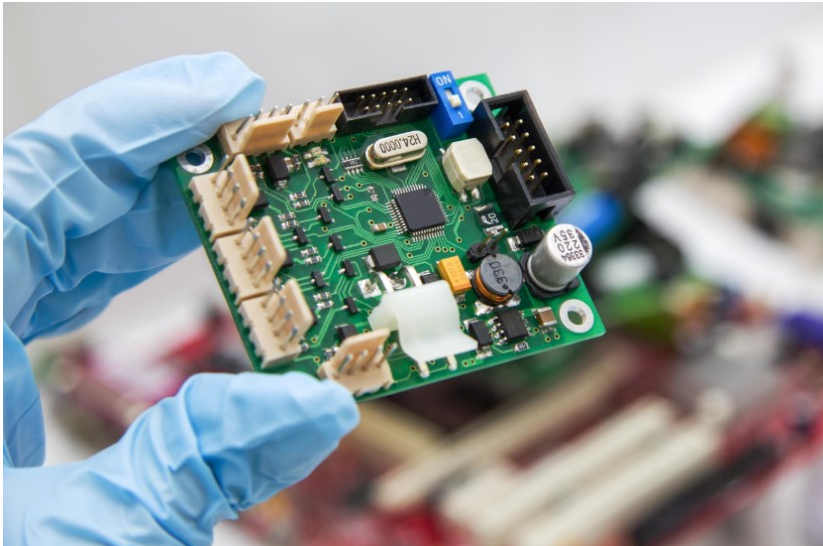
Integrated Circuit (IC) Design



Chips are produced on wafers, and a single wafer can house numerous chips

A single chip cut from a wafer is covered in a resin package to become a semiconductor product

PCB Design



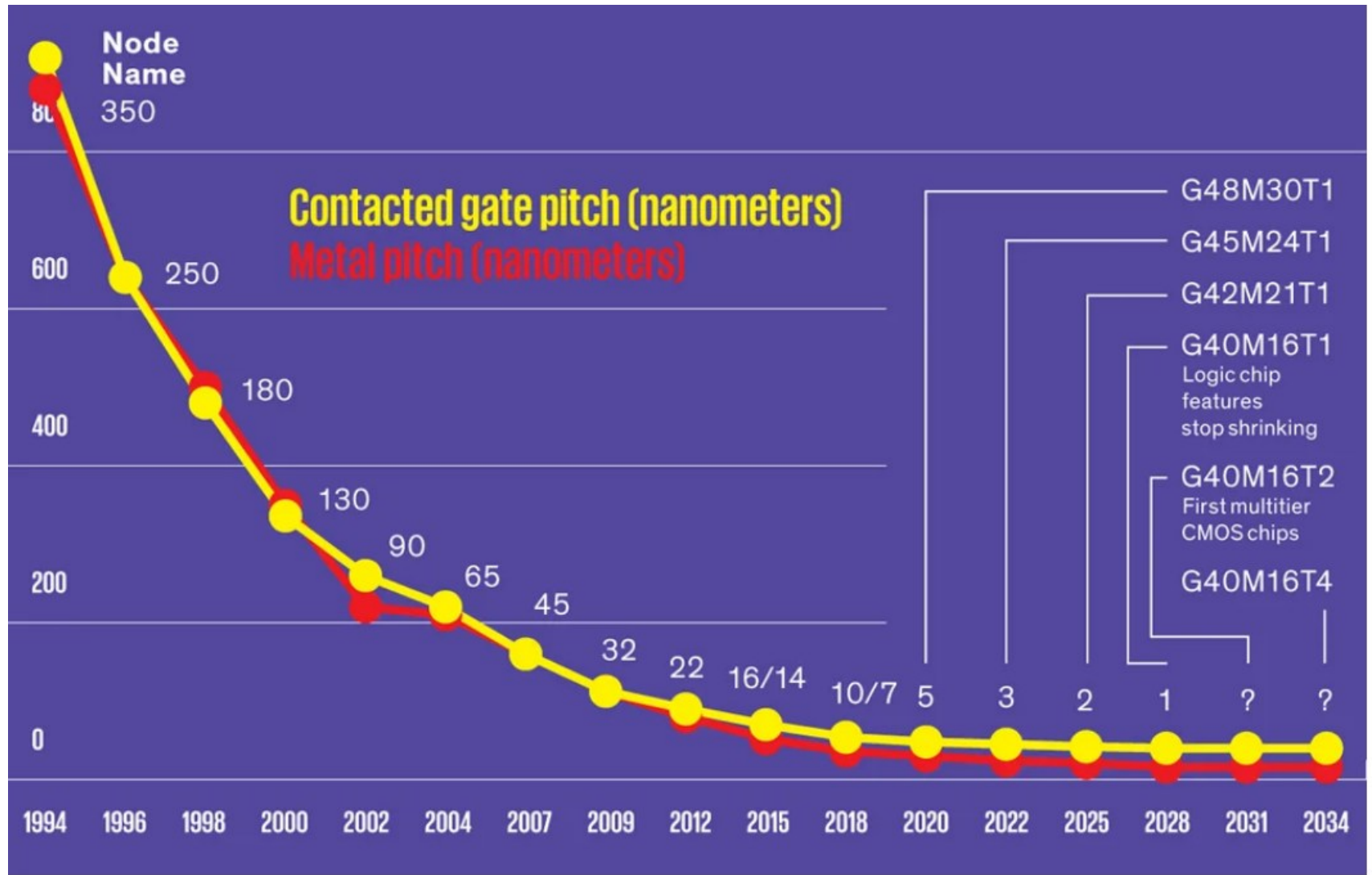
Integrated Circuit (IC) Design



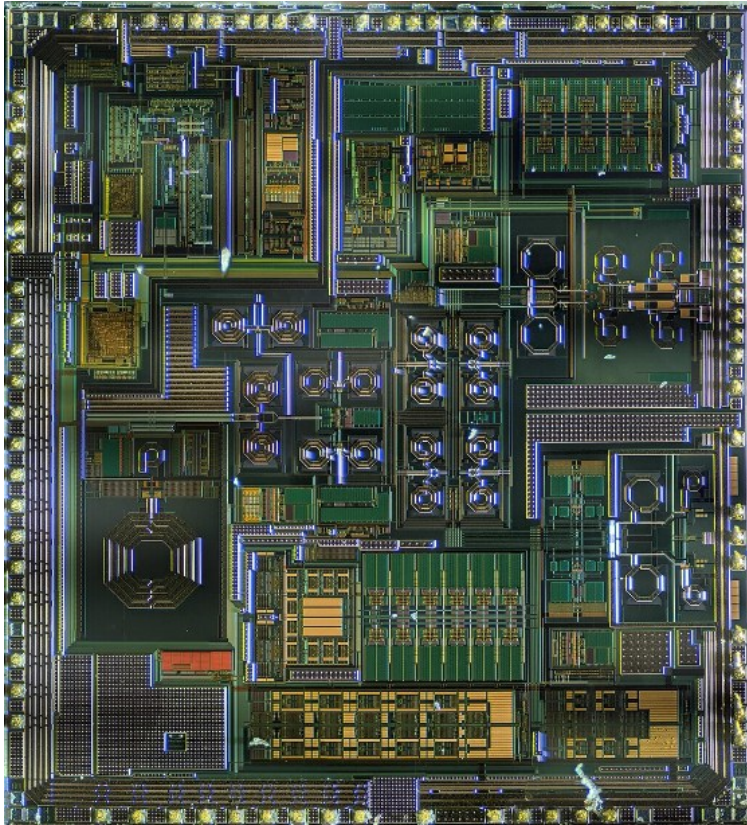
Integrated Circuit (IC) Design



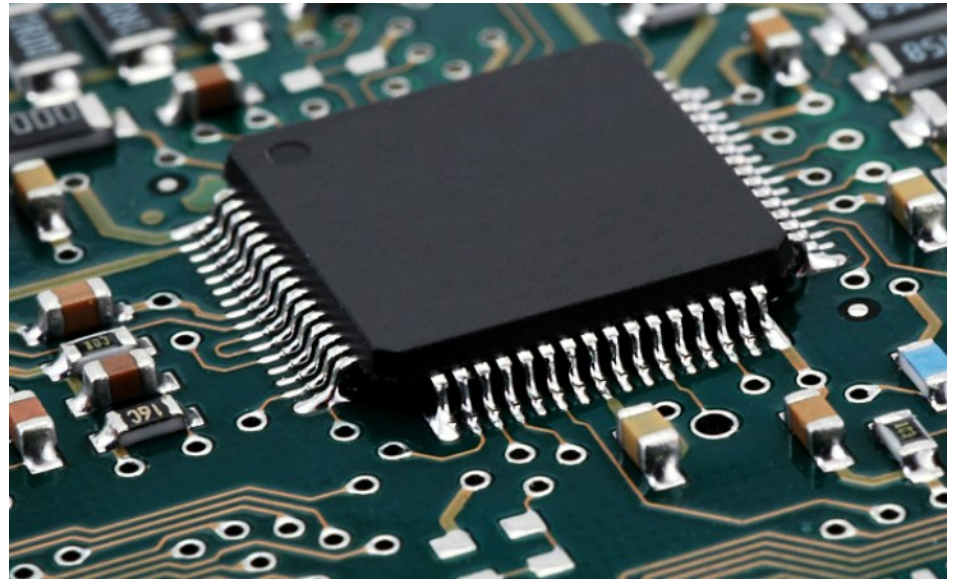
Transistors



SOURCES: STANFORD NANOELECTRONICS LAB, IEEE INTERNATIONAL ROADMAP FOR DEVICES AND SYSTEMS 2020

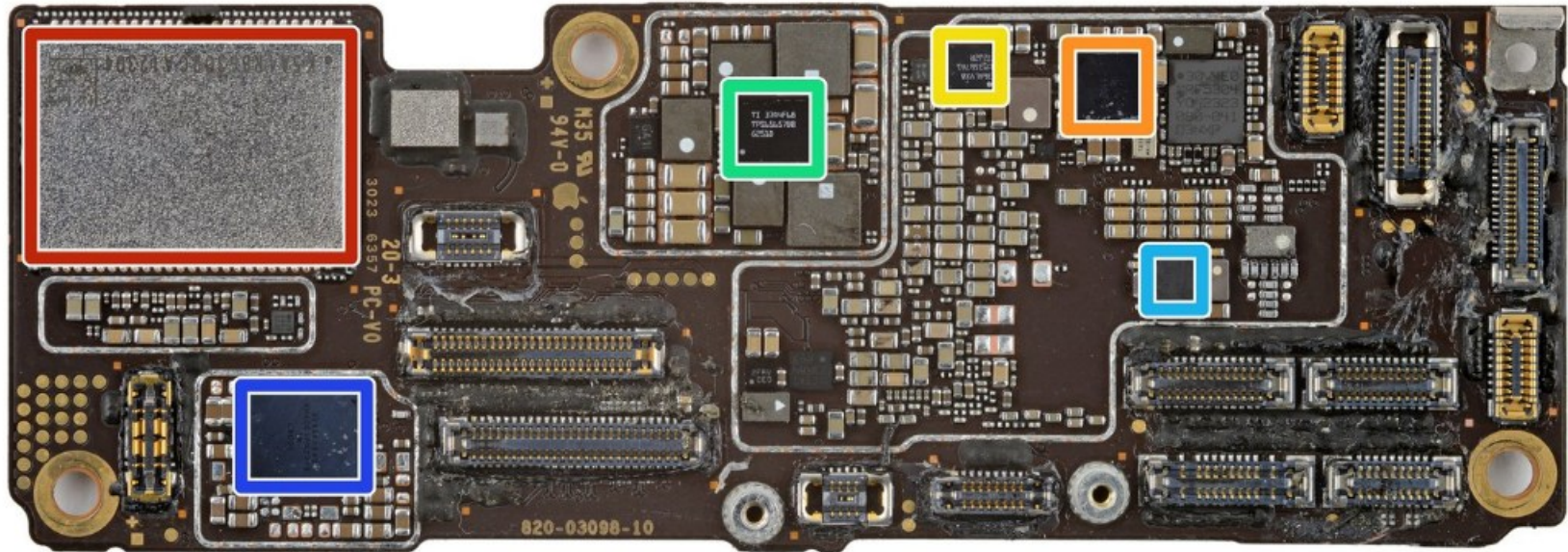


Integrated Circuit Design



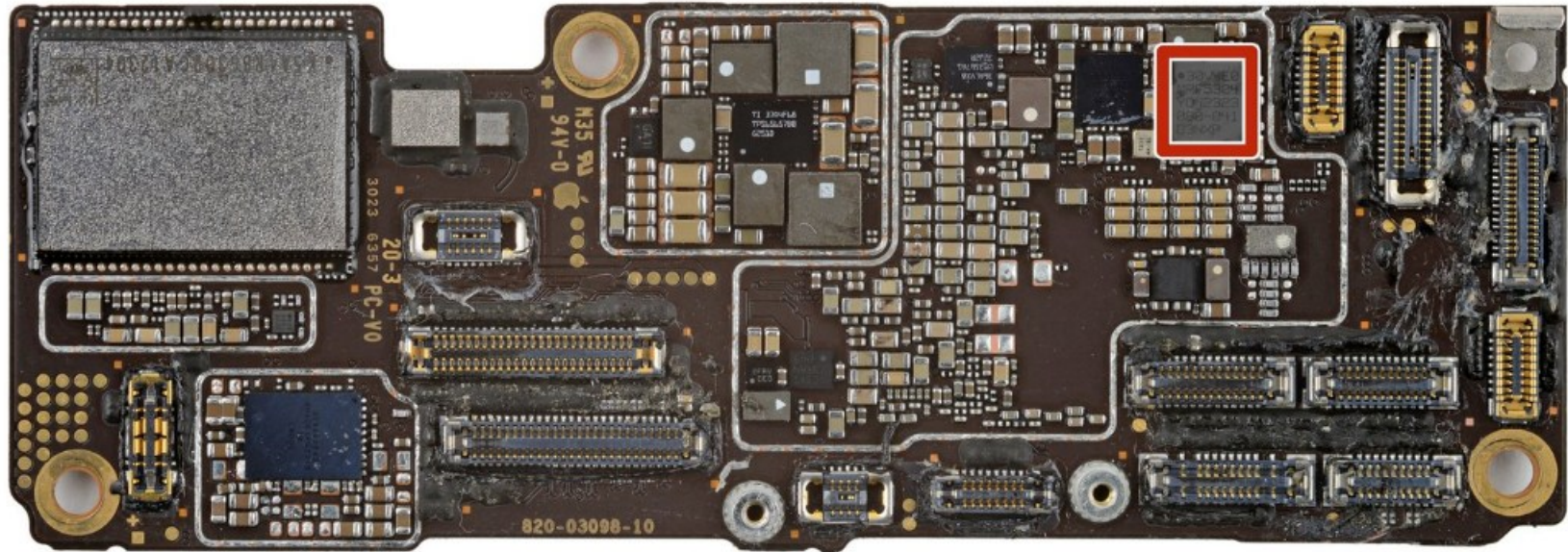
PCB Design

Telecommunications - IPHONE 15 Pro Max



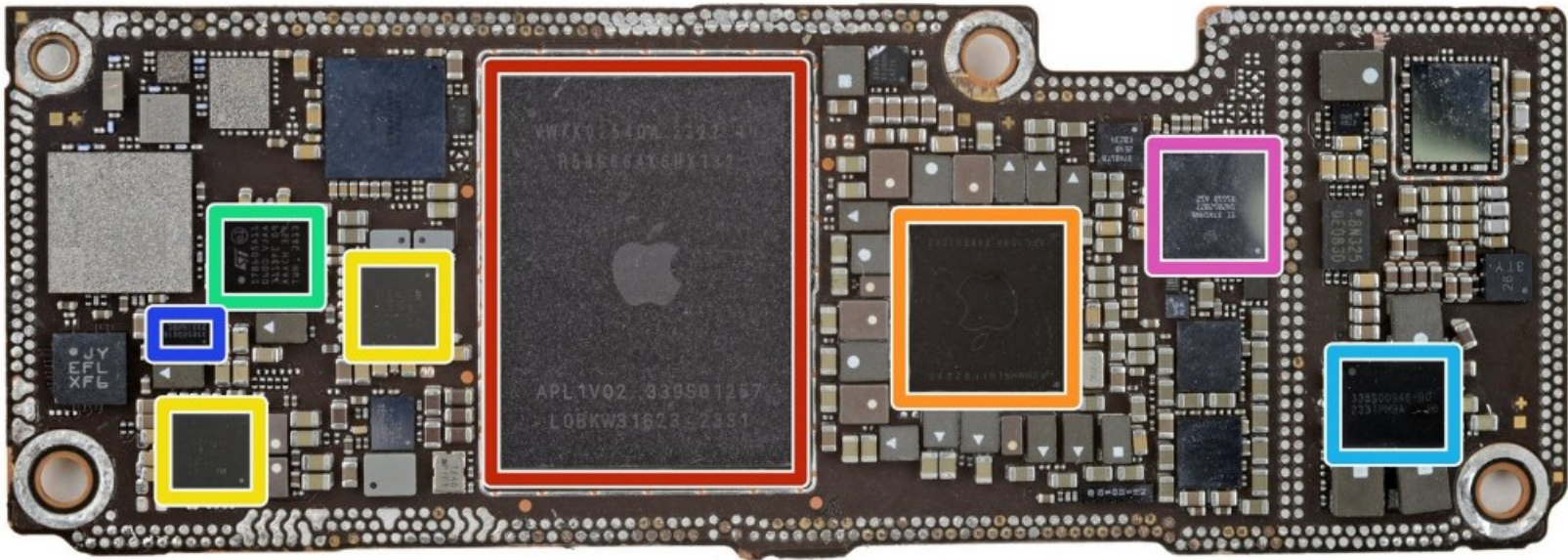
- Likely Kioxia K5A4RB6302CA12304 256 GB NAND flash
- Apple 338S00537 audio amplifier
- Texas Instruments LM3567A1 flash controller
- Texas Instruments TPS65657B0 display power supply
- Apple 338S01026-B1 power management
- Possibly Apple 338S00843 audio DSP

Telecommunications - IPHONE 15 Pro Max



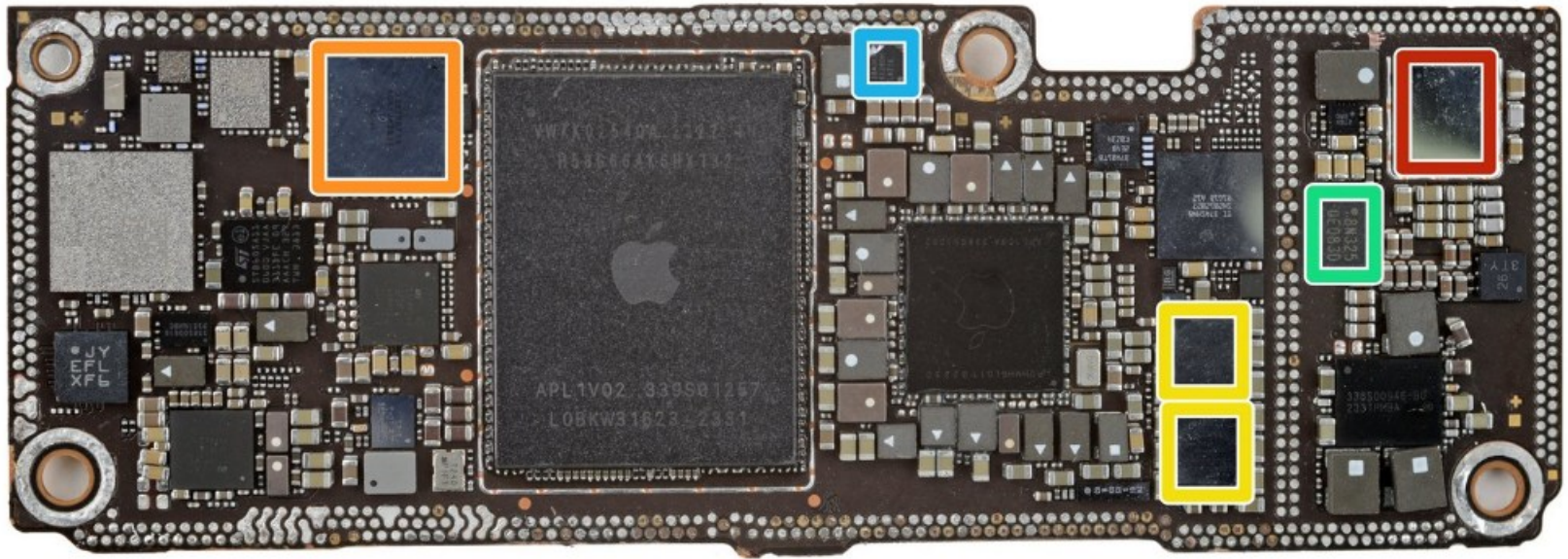
- Likely NXP Semiconductor NFC controller

Telecommunications - IPHONE 15 Pro Max



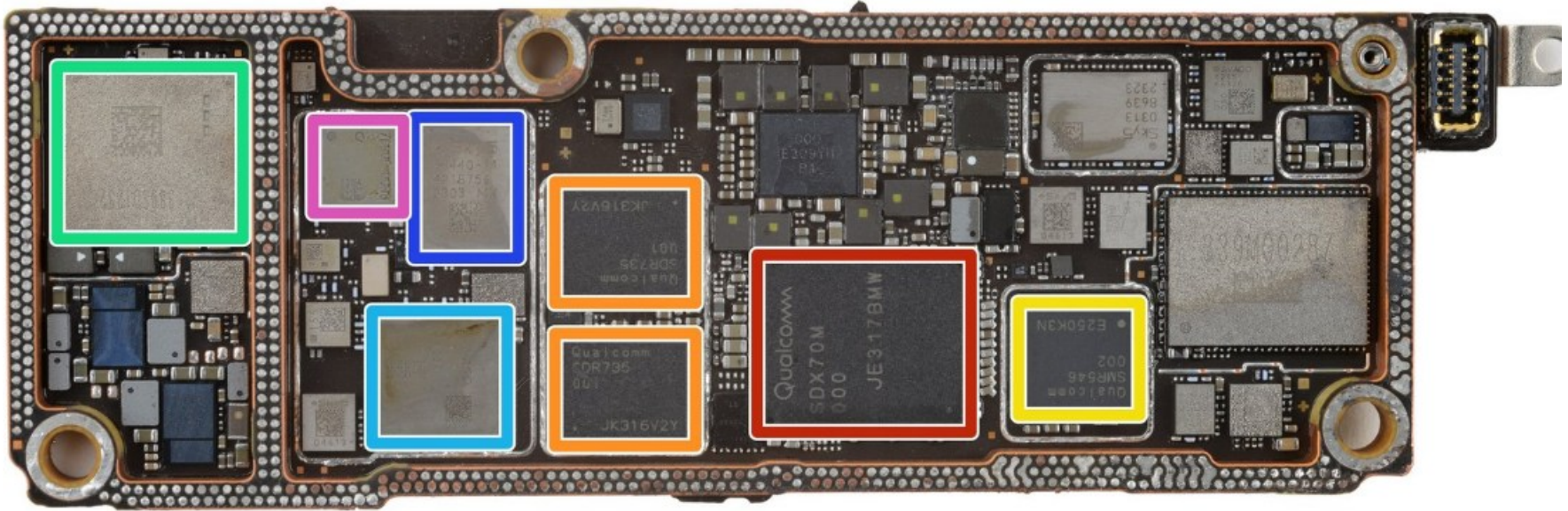
- Apple APL1V02/339S01257 A17 Pro hexa-core applications processor w/ hexa-core GPU layered under likely a SK hynix H58G66AK6HX132 8 GB LPDDR5 SDRAM memory
- Apple APL109A/338S01022 power management
- STMicroelectronics STCPM1A3 power management
- STMicroelectronics STB605A11 power management
- Apple 338S00946-B0 power management
- Apple 338S00616 power management
- Likely Texas Instruments SN2012017 battery charger

Telecommunications - IPHONE 15 Pro Max



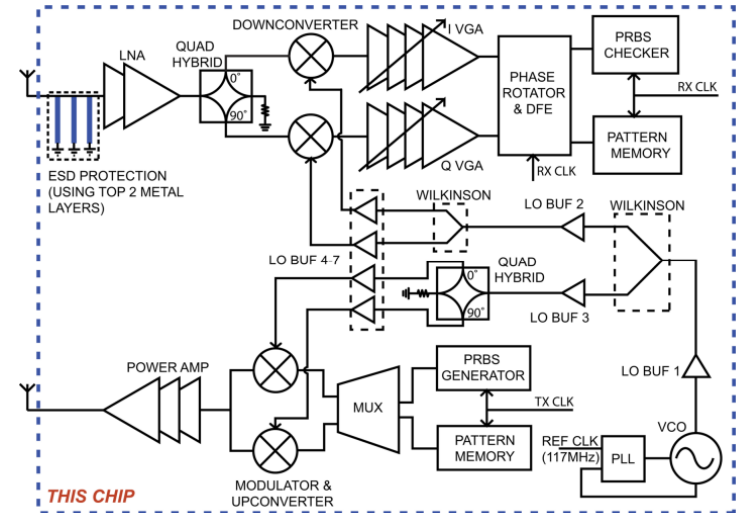
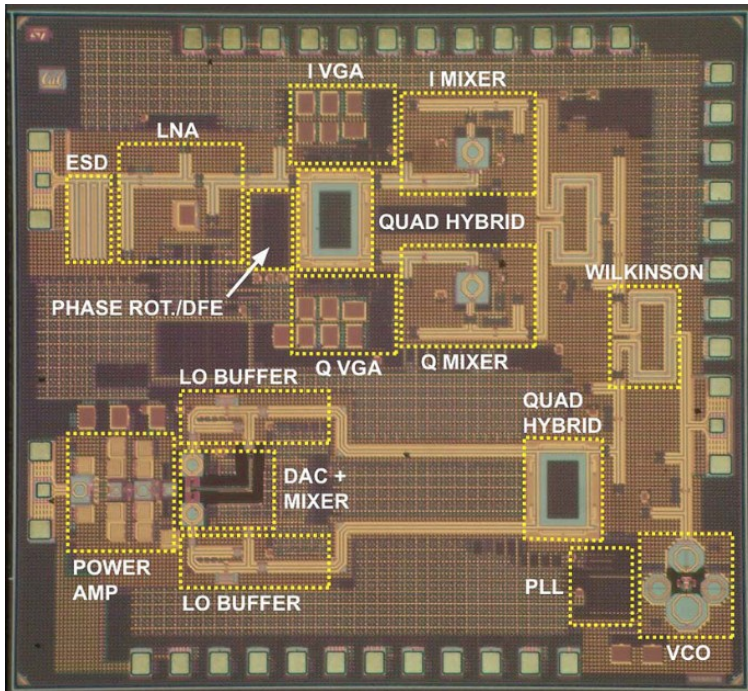
- Broadcom BCM59365EA1IUBG wireless power receiver
- Apple 338S00739 audio CODEC
- Apple 338S00537 audio amplifier
- Winbond [W25Q80DVUXIE](#) 1 MB serial NOR flash memory
- Likely Texas Instruments TPS61280H battery front-end DC-DC converter

Telecommunications - IPHONE 15 Pro Max



- Qualcomm [SDX70M](#) Snapdragon X70 modem
- Qualcomm SDR735 RF transceiver
- Qualcomm SMR546 RF transceiver
- Possibly Apple 339501232 WiFi & Bluetooth module
- Broadcom AFEM-8234 front-end module
- Skyworks SKY58440-11 front-end module
- Qorvo QM76305 front-end module

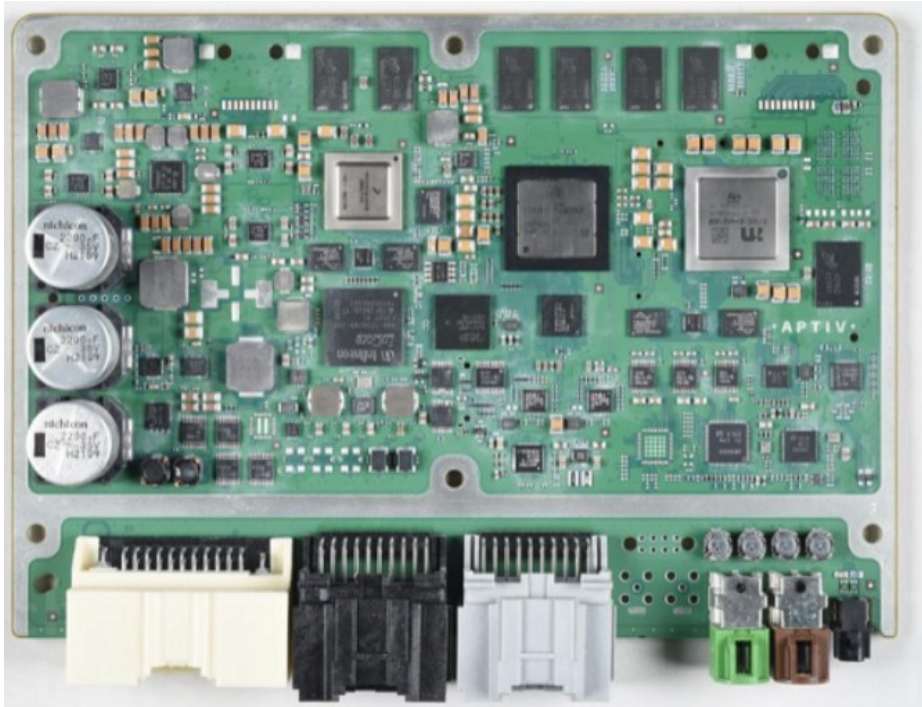
Telecommunications Radio Frequency Transceiver Chip Photo



Block Diagram of RF Transceiver

90 nm CMOS – Chip size: 2.5 x 2.75 mm²

Automotive – Electronic Control Unit

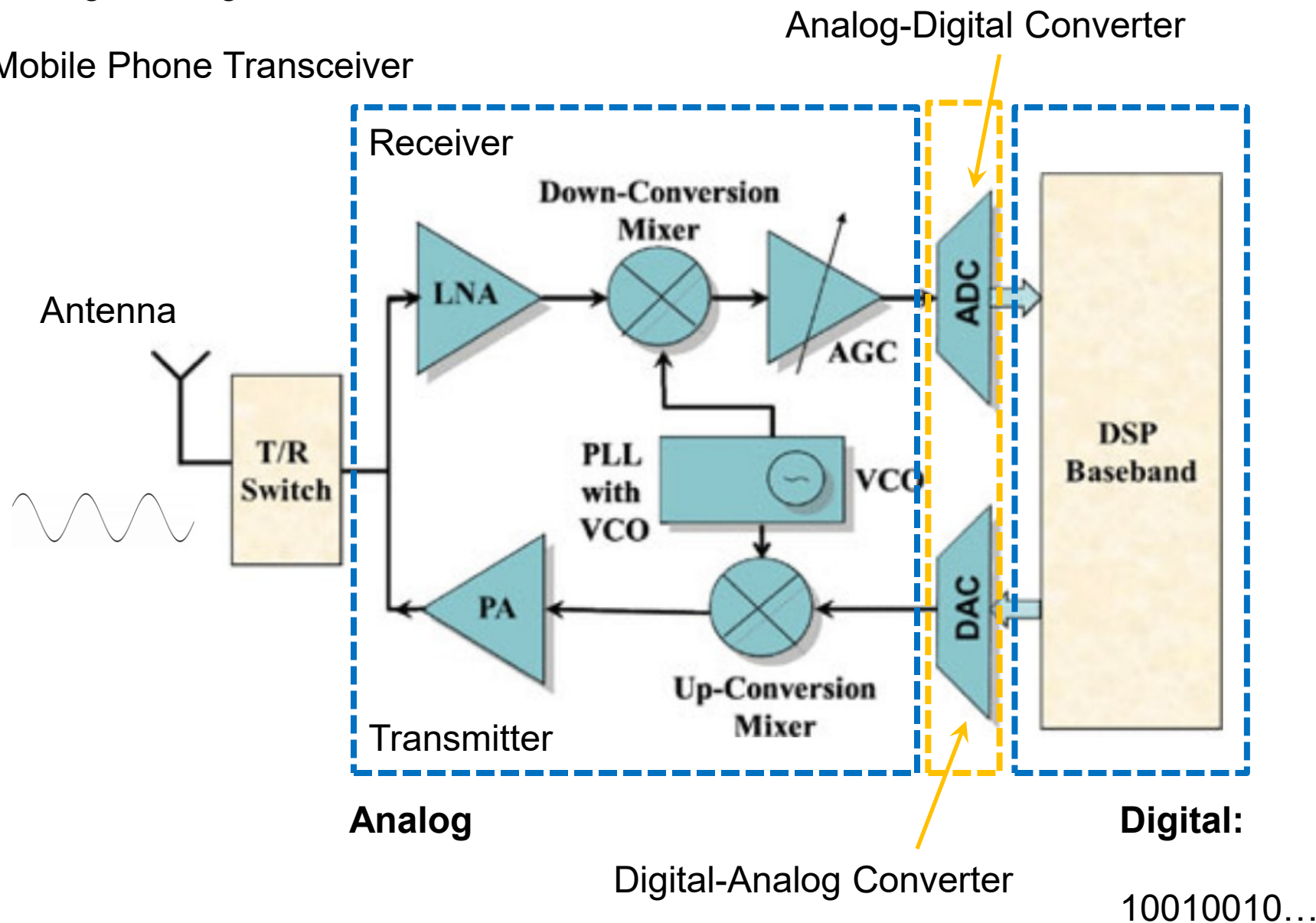


Ford F150 Electronic Control Unit



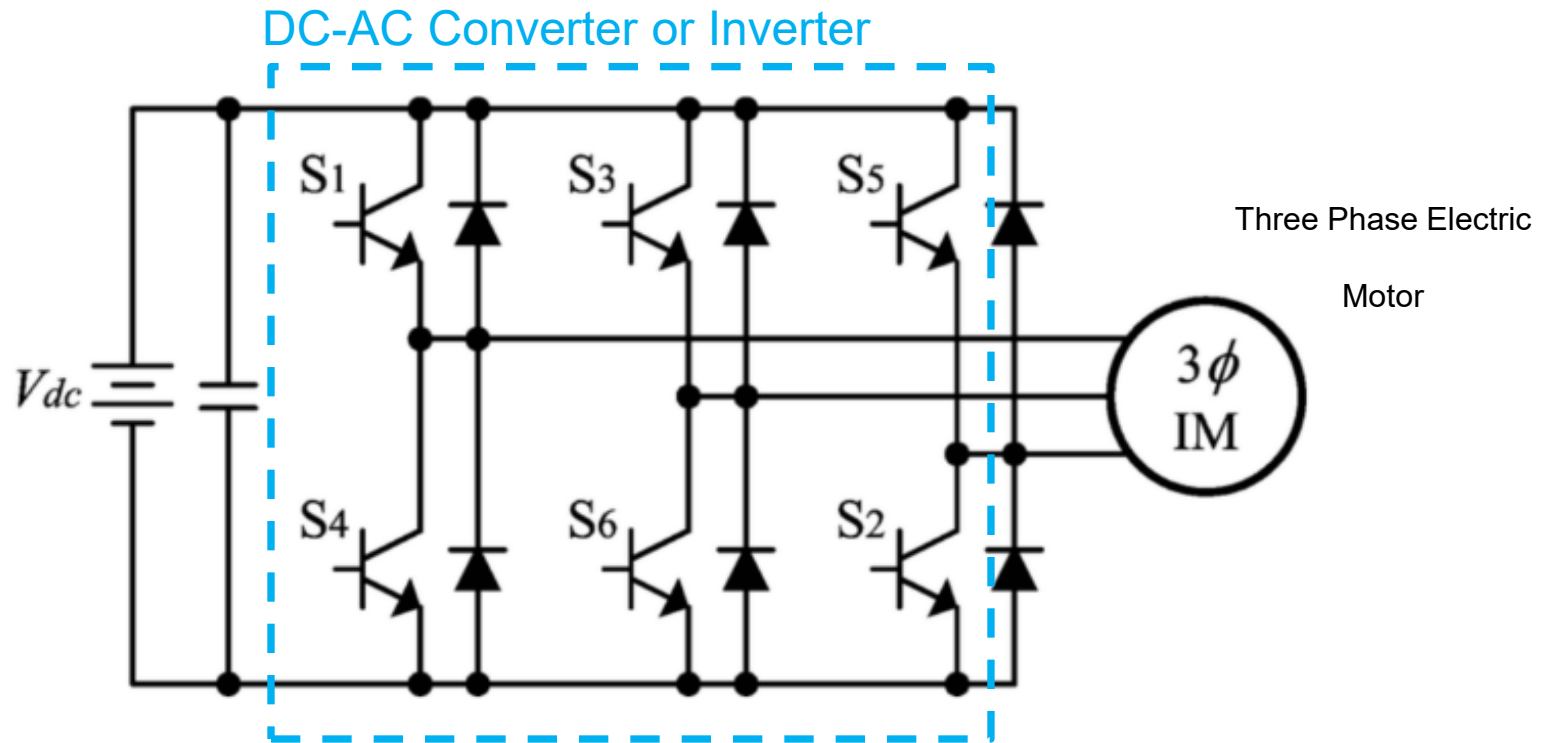
Analog and Digital Circuits

Mobile Phone Transceiver



Analog and Digital Circuits

Analog Circuit Example

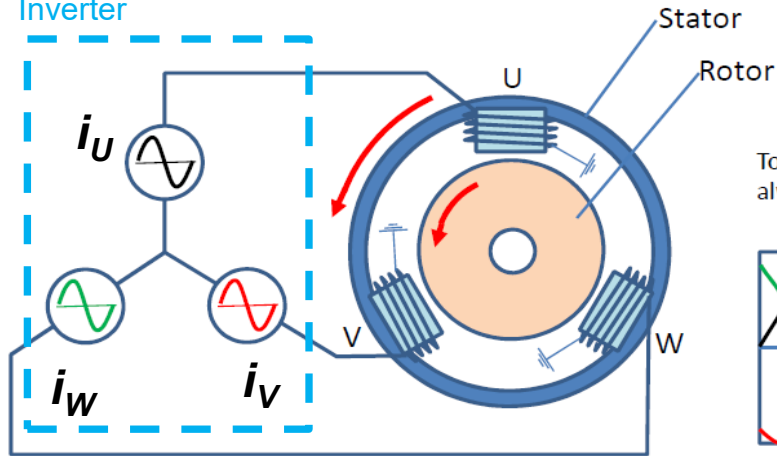


Three Phase Inverter: Electric Vehicles, Electric Pumps etc.

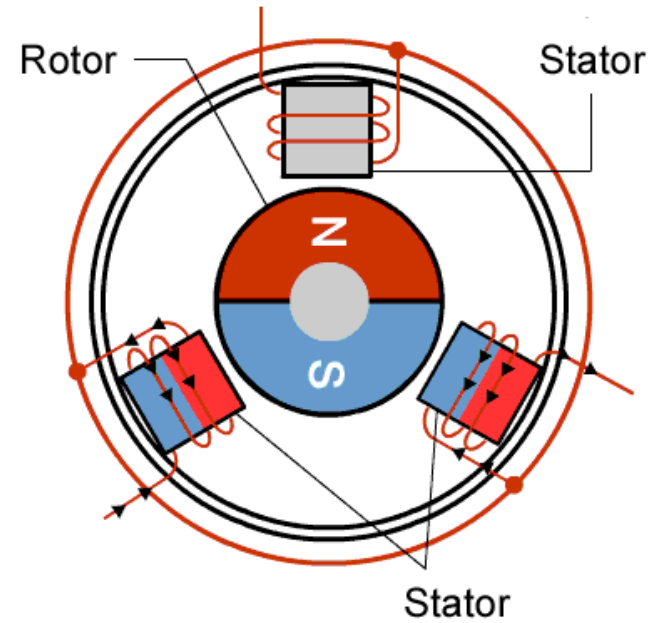
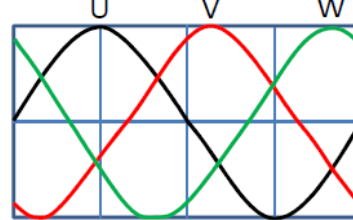
Analog and Digital Circuits

Analog Circuit Example

Symbolic Representation of Inverter



Total value of U, V, W phase currents is always kept zero.



Three Phase System

120° Phase Difference between each Phase

A current is injected into each winding (copper wire) of the electric motor with a 120° phase shift. This current generates an electromagnetic field/force (EMF) which repels or pulls the rotor depending on the rotor position and the stator current.

Analog and Digital Circuits

Digital Circuits:

- Processors
- FPGAs
- Logic Gates
- Flash Memory
- RAM
- EEPROM
- ...



Processor



FPGA



LPDDR4 RAM

Analog and Digital Circuits

Mixed Signal Design: a combination of both analog and digital circuit design

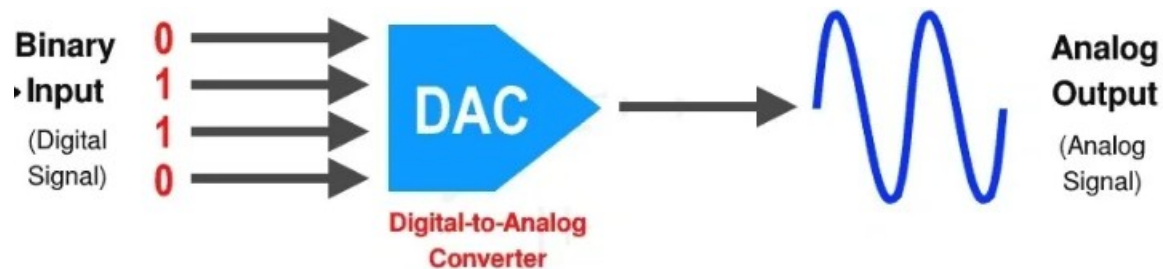
ADC: Analog-Digital Converter

An ADC converts an analog signal to a digital one.



DAC: Analog-Digital Converter

A DAC converts a digital signal to an analog one.



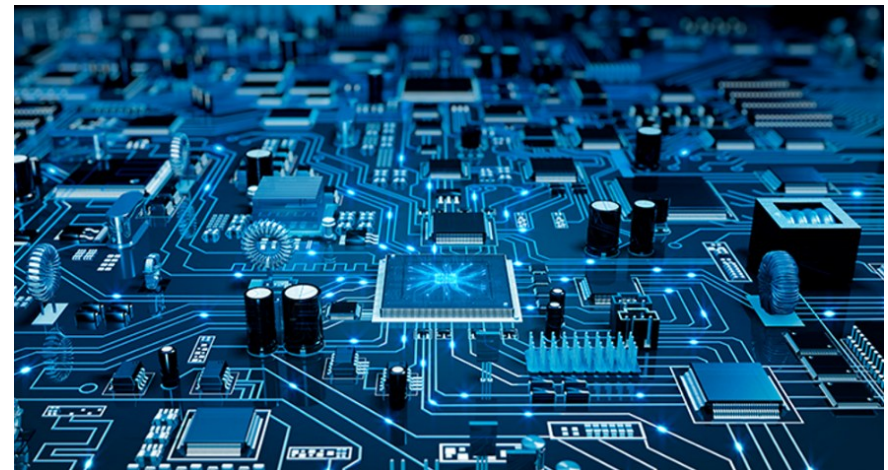
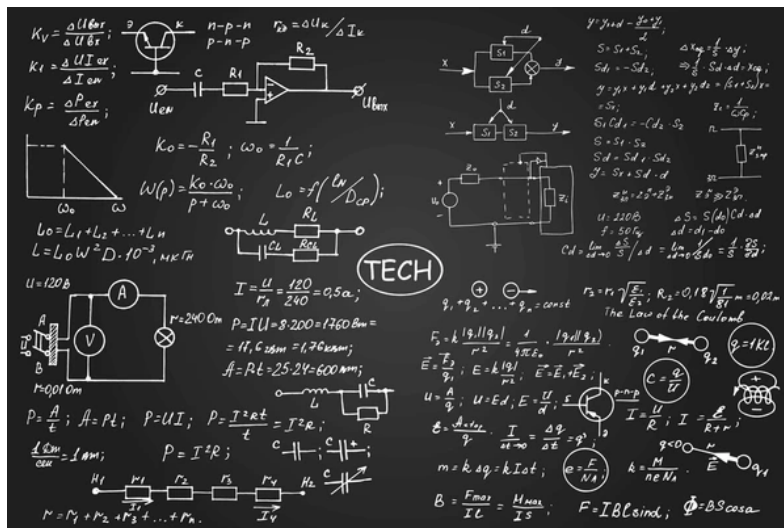
Electronic hardware design can be done at both PCB and chip level.

Electronic hardware circuits range from power supply circuits, microcontrollers, sensors to communication interfaces (USB, RF, ethernet etc.).

The design of these complex circuits implies a solid understanding and a good theoretical background of electronic circuits!

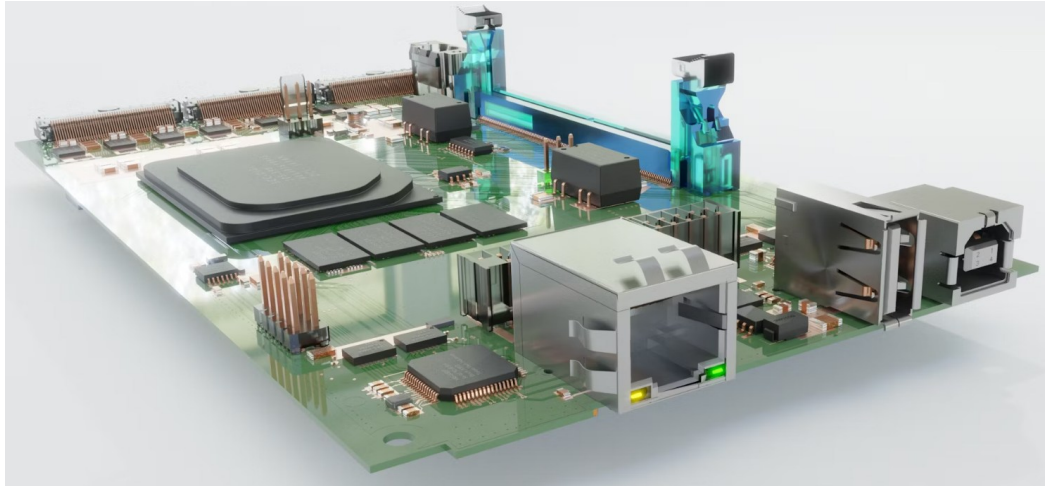
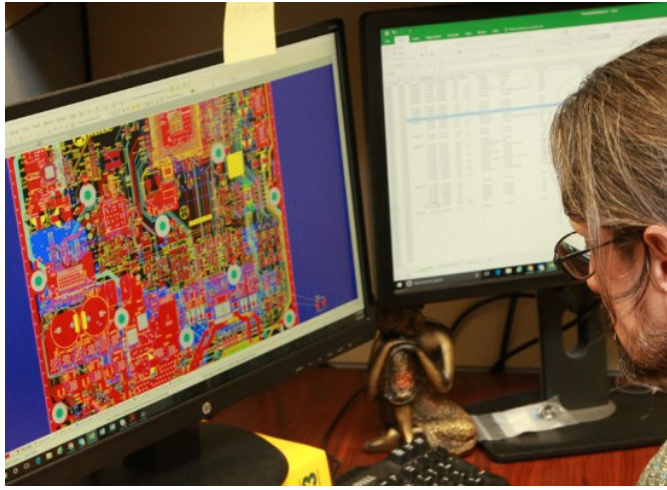
We will start with basic circuit component components like resistors, capacitors, inductors. We will also talk about the fundamentals of amplifiers, however not at transistor level! Before we start with the design of complex circuits at transistor level, we need to understand the fundamentals of electronic circuits within the scope of this lecture...

There is a huge demand for good electronic hardware and embedded design (hardware & software) engineers from defence, automotive, medical and white goods sectors!



Throughout this course, CMOS transistors will be given more emphasis compared to BJTs because CMOS technology is more dominant in today's technology!

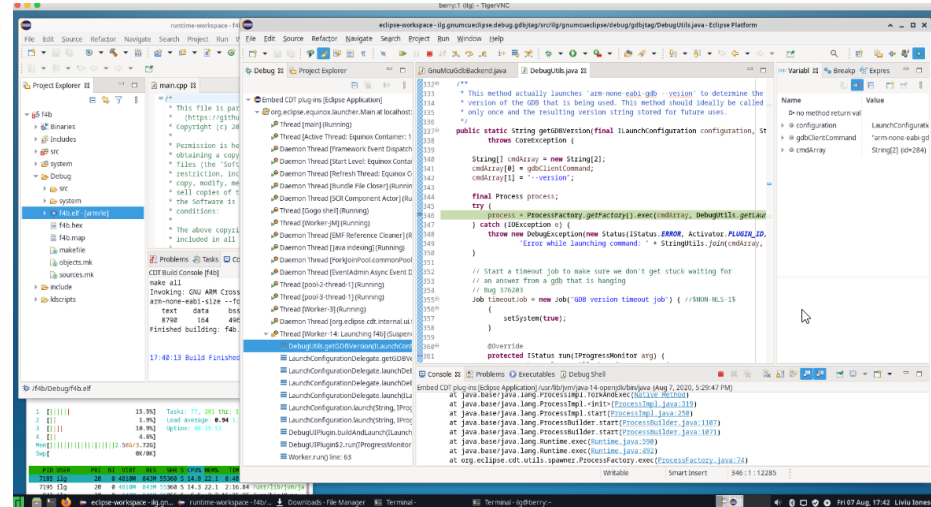
What does an electrical and electronics engineer do?



Electronic Hardware Design

- White Goods and Consumer Products
- Medical
- Automotive
- Defence
- Industrial
- Telecommunications
- IoT

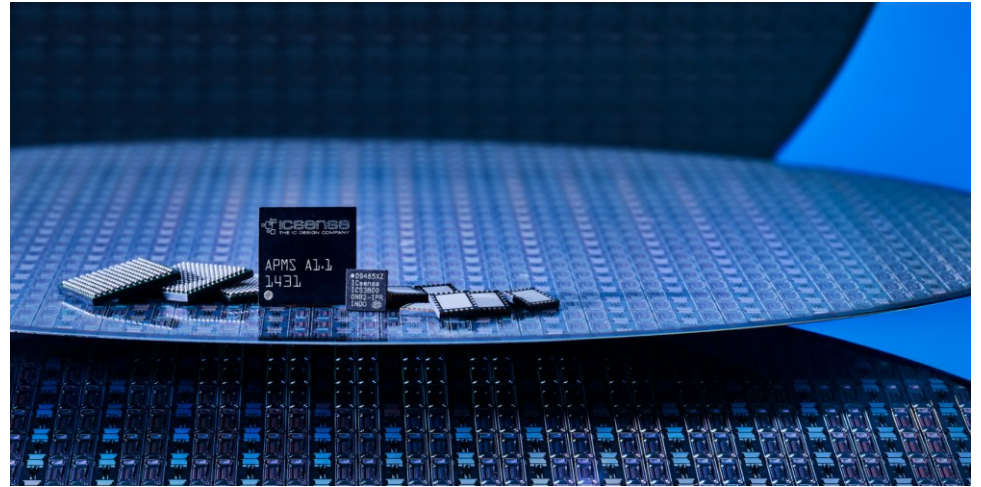
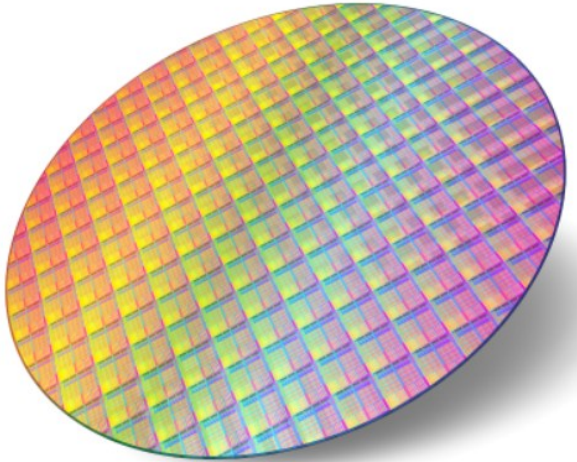
What does an electrical and electronics engineer do?



Embedded Software Design

- White Goods and Consumer Products
- Medical
- Automotive
- Defence
- Industrial
- Telecommunications
- IoT

What does an electrical and electronics engineer do?



Integrated Circuit Design

- Medical
- Automotive
- Defence
- Industrial
- Telecommunications
- IoT

What does an electrical and electronics engineer do?



Test Engineering (SW & HW)

- Medical
- Automotive
- Defence
- Industrial
- Telecommunications
- IoT

What does an electrical and electronics engineer do?

Main topics are

- IC Design: analog, digital, mixed-signal (ICs are used in nearly every application)
- Electronic Hardware PCB Design: electronic control units, communication units (IoT, mobile phones, base stations, PCs...)
- Embedded Software: Control Applications (sensor and actuator), Cybersecurity, Telecommunications (2G, 2.5G, 3G, 4G, 5G, WiFi, Bluetooth, CAN, USB, SPI, I2C...) both for mobile phones and base stations. Embedded software is necessary for every application that involves a microcontroller, DSP, FPGA, microprocessor etc.
- Test Engineering: Test of electronic hardware, (embedded software), programming of graphical user interfaces (GUI)
- ...